

RIANA MALHOTRA

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EDUCATION

Bachelor of Applied Science - Computer Engineering + PEY Co-op

Expected May 2029

University of Toronto

- UofT Engineering International Scholar Award Recipient
- Minors: Artificial Intelligence and Engineering Business

TECHNICAL SKILLS

Programming: C++, C, Python, C#, Java, Verilog, Assembly, SQL, MATLAB

Systems & Architecture: Linux/UNIX, QNX RTOS, Computer Architecture, Multithreading, Hardware/Software Integration

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Debugging, Software Testing, Regression Testing, Performance Optimization

AI & Machine Learning: Machine Learning, Generative AI, PyTorch, NumPy, Computer Vision, Image Processing

Hardware & Development: RTL Design, FPGA, Digital Circuits, Microcontrollers, UART, CAN, GPIO, Timers, Git

EXPERIENCE

Production Test Engineer Intern | Trimble Applanix

May 2025 - Aug 2025, May 2026 - Aug 2026

- Developed and debugged 15+ C# and C++ tools for firmware programming, device configuration, hardware communication, and diagnostics, alongside Python automation used across 50+ production builds, reducing programming time by approximately 20%.
- Developed and executed automated integration, regression, and system tests for GNSS/INS firmware in Linux and QNX environments, analyzing logs, communication interfaces, and runtime behavior to identify failures across software and hardware.
- Investigated firmware and system failures by reproducing defects, analyzing runtime behavior and test results, and collaborating with software, firmware, QA, and production teams to isolate root causes and verify fixes.

Software Engineer | Outlier.ai

Jan 2025 - May 2026

- Trained and evaluated large language models (LLMs) on software engineering tasks including algorithmic problem solving, code generation, debugging, and data-structure reasoning.
- Designed structured test cases with expected outputs and edge cases to evaluate generated Python, C++, and Java code for functional correctness, robustness, and adherence to requirements.
- Performed systematic error analysis of generated code, identifying algorithmic and implementation bugs and analyzing failure patterns to improve training data and model responses.

Software Engineering Member - Autonomous Rover Team | University of Toronto Robotics Association

Jan 2024 - Dec 2024

- Contributed C++ and Python software for autonomous navigation, sensor processing, and real-time data processing as part of the rover autonomy system.
- Developed Python testing and diagnostic tools to evaluate system behavior, reproduce failures, and validate software changes across simulation and hardware.
- Collaborated with autonomy and hardware teams to integrate software on the rover, analyze test results, and debug system-level issues through repeated trials.

PROJECTS

TelMe - GenAI Telecom Support Platform | ML & Backend Developer | Python, Flask, React, Node.js, REST APIs, Machine Learning

- Built a full-stack AI application during a 24-hour hackathon, developing Python backend services and REST APIs for AI-assisted customer support.
- Implemented K-means clustering in Python to analyze customer usage patterns, group similar users, and generate personalized telecom plan recommendations.
- Tested end-to-end workflows and analyzed application behavior, debugging integration issues across the Python, React, and Node.js components while rapidly iterating toward a working prototype.

TransitFlow - GIS for Commuters | Software Lead | C++, Data Structures & Algorithms, Multithreading, Performance Optimization

- Designed and developed a C++ GIS application for commuter route planning using A*, Dijkstra, Greedy, Ant Colony Optimization, 2-opt, OR-opt, and Simulated Annealing.
- Built a multithreaded optimization pipeline and evaluated algorithm performance across different scenarios, improving solution quality by 8% through algorithm optimization and performance testing.
- Developed test cases and measured runtime and solution quality to identify defects, compare implementations, and validate performance across workloads.

FPGA DJ Booth - Embedded Audio Processing System | Embedded Systems Developer | FPGA, Nios V, Assembly, Digital Design

- Built an FPGA-based audio processing system using the Nios V processor, integrating processor software, memory, peripherals, microphone input, and VGA output for real-time processing.
- Implemented real-time DSP functionality for echo, delay, reverb, and volume control while managing processor-peripheral communication and system timing.
- Debugged hardware/software interactions by tracing memory behavior, peripheral communication, timing, and runtime audio behavior on FPGA hardware.